MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

THIRD SEMESTER PRACTICAL EXAMINATION

(2024 ADMISION ONWARDS)

MG3DSCECT204 DIGITAL FUNDAMENTALS

Duration: 1.5 hours Maximum Marks: 35

Practical Examination – Model Question Paper

Instructions:

1. Demonstrate the given task.
2. A Viva session will follow demonstration.
   (Minimum 3 questions from the Practical Module)
3. Lab Report should cover all the subunits of the practical module.

Evaluation Criteria:

- **Viva:** 7 marks
- **Lab Report:** 8 marks
- **Demonstration:** 20 marks

Tasks: *(Choose any one from the list)*

1. Verify the truth table of basic logic gates with necessary diagrams.
2. Implement a basic Boolean equation and verify the truth table.
3. Verify the De Morgan's theorem with the help of a truth table.
4. Verify the universal property of NAND gate.
5. Verify the universal property of NOR gate.
6. Verify the truth table of RS flip flop.
7. Verify the truth table of JK flip flop.
8. Implement a half adder circuit and verify the truth table.
9. Implement a full adder circuit and verify the truth table.
10. Implement a basic multiplexer circuit.
11. Implement a basic de multiplexer circuit.
12. Implement a serial in serial out shift register.
13. Implement a serial in parallel out shift register.